Expert Systems With Applications 185 (2021) 115656

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

Bias and variance residuals for machine learning nonlinear simplex regressions

Patrícia L. Espinheira *, Luana C.M. Silva, Francisco Cribari-Neto

Departamento de Estatística, Universidade Federal de Pernambuco, Cidade Universitária, Recife/PE, 50670-901, Brazil

ARTICLE INFO

Keywords:
Double bounded variable
Machine learning
Outlier
Simplex regression

ABSTRACT

We propose two new residuals that can be used to evaluate the bias and variance of nonlinear simplex regressions for machine learning. Such models are supervised learning tools for problems with high complexity, since they involve nonlinearity, and are used with univariate responses that assume values in the standard unit interval. The residuals we introduce are obtained from Fisher's iterative scoring algorithm used for estimating the mean and dispersion regression coefficients. A novel feature of the proposed residuals is that they do not require the computation of projection matrices. As is well known, the computation of such matrices can be computationally costly when the sample size is large. We present and discuss three empirical applications, two that use real data and one that is based on a simulated dataset. The results from all three empirical analyses favor the proposed residuals.

1. Introduction

Many machine learning techniques have been unsuccessful because their classifications are (a) directly obtained from the raw data without feature extraction or (b) obtained from data contaminated with outliers or (c) performed without adding replacement values for missing values or (d) obtained after filling missing values simply with the mean value; see Maniruzzaman et al. (2018). It is then clear that outlying data points may negatively impact machine learning techniques and that improvements can be achieved by properly identifying such data points and accounting for them; see, e.g., Weizhi et al. (2015). One of our chief interests in this paper relates to outlier identification in the context of supervised machine learning in regression analyses in which the variable of interest assumes values in the standard unit interval, $(0, 1)$. In particular, we seek to identify data points that disproportionately impact the estimates of the parameters that index the model. As noted by Belhadi et al. (2021) in the context of machine learning for trajectory data, it is important to not only detect individual outliers but also outliers that appear in groups. The diagnostic analysis we present for supervised machine learning can be used to identify individual outliers and also groups of outlying data points. We shall refer to (individual and group) outlying cases as influential observations due to the impact they exert on the model fit.

Outlier detection is closely related to data mining since it is performed via residual analyses that aim at uncovering data anomalies, patterns and correlations that can assist decision making processes. According to Han et al. (2011), data mining tasks that employ supervised learning typically aim at producing predictions about the phenomenon under study. In that context, regression analysis relates to modeling, evaluating, and distributing information. Indeed, regression models are commonly used for supervised machine learning (Weiss & Indurkhya, 1997). Data mining modeling uses algorithms that seek to identify how the predictor attributes (covariates in regression) are associated with the class (response in regression). The model essentially represents knowledge on the classification, a relationship between values of the predictor attributes (covariates) and the class (response), such that, given a set of values for the predictor attributes, prediction of the class can be performed. There is also interest in mining patterns that must be incorporated into the model so that the resulting predictions are as accurate as possible. A simple data mining algorithm based on the linear regression model was proposed by Gupta (2015). In what follows, our interest will not lie in such a model, but in regression models tailored for double-bounded variables.

The most critical step of the mining process is the evaluation step, which aims at determining whether any aspects important to the problem have been overlooked. Data points that are considered "abnormal" (outliers) and patterns behind the emergence of these data points are mined. Finally, in the distribution step, the knowledge gathered is organized so that it can be shared. Residual analysis, as we shall see later, is used in data mining modeling tasks and in the validation of such modeling. The performance of a residual analysis depends heavily on the quality of the residual used, that is, on its ability to mine patterns

* Corresponding author.
E-mail addresses: patespipa@de.ufpe.br (P.L. Espinheira), lcmds1@de.ufpe.br (L.C.M. Silva), cribari@de.ufpe.br (F. Cribari-Neto).

https://doi.org/10.1016/j.eswa.2021.115656
Received 14 May 2020; Received in revised form 4 March 2021; Accepted 22 July 2021
Available online 30 July 2021

and anomalous cases in the data. In the present paper, we propose two residuals that typically outperform, as we will see, existing residuals for an important model used with double-bounded data: the simplex regression model.

Models for double-bounded dependent variables that assume values in the standard unit interval have been receiving increasing attention in the literature. Different regression models are tailored to such responses, e.g.: the simplex (Barndorff-Nielsen & Jorgensen, 1991), beta (Ferrari & Cribari-Neto, 2004), Kumaraswamy (Mitnik & Baek, 2013), Johnson S_B (Lemonte & Bazán, 2015), unit gamma (Mousa et al., 2016) and CDF-quantile (Smithson & Shou, 2017) regression models, the latter being based on a two-parameter family of distributions which is obtained by applying the cumulative distribution function of a parent law to the quantile function of another distribution. It is noteworthy that the simplex model belongs to the class of dispersion models (Jørgensen, 1997) which extends the well known class of generalized linear regression models. In its most basic formulation, dispersion is taken to be constant. A more general formulation consists of two submodels, one for the response mean and another one for the dispersion. Song and Tan (2000) developed a constant dispersion marginal model for longitudinal double-bounded variables and developed parameter estimation via generalized estimating equations. A generalized version of their model in which correlations and dispersion effects are modeled together with mean effects was considered by Song et al. (2004). Qiu et al. (2008) introduced the mixed effects simplex model coupled with bias-corrected penalized quasi-likelihood estimation. A review of dispersion models and of maximum likelihood inference for such models can be found in Song (2009). More recently, Zhang et al. (2016) developed the `simplexreg` package for the R statistical computing environment; it can be obtained from the Comprehensive R Archive Network (CRAN). The class of nonlinear simplex regression models was proposed by Espinheira and Silva (2020). The authors defined the model, presented closed-form expressions for the score function and for Fisher's information matrix, outlined maximum likelihood parameter estimation, and presented some diagnostic measures, such as a residual and local influence measures. A novel feature of the simplex model is that, unlike competing models, it can be used with bimodal data. Additionally, it typically outperforms the beta model when the response values are close to the lower or upper standard unit interval limit in the sense that its parameter estimates are less sensitive to the outlying data points.

Diagnostic analysis comprises of a set of procedures that can be used to validate a fitted regression model. The conclusions drawn by the practitioner on how the covariates impact the response are based on a postulated model, and it is thus important to determine whether the model at hand fits the data well. That can be accomplished by residual analyses, which are used to identify discrepancies between the fitted model and the observed data. Different diagnostic tools were developed for the beta regression model and also for some alternative models. For instance, Espinheira et al. (2008) developed local influence analysis for beta regressions and Lemonte and Bazán (2015) obtained similar results for the Johnson S_B regression model. A residual tailored to simplex regressions was obtained by Espinheira and Silva (2020) who also developed local influence analyses based on different perturbation schemes. Espinheira et al. (2019) proposed a model selection criteria based on computed residuals for beta regression for machine learning that can be used to detect model overfitting and model misspecification.

Trade-offs between bias and variance are key in machine learning processes. Our chief goal in this article is to develop residuals that can be used to assess the bias and variance of the nonlinear simplex regression learning. In particular, we develop two new residuals ('variance residual' and 'bias variance residual') for use with such models. They are obtained from Fisher's iterative scoring scheme for maximum likelihood parameter estimation. When deriving the new residuals we consider joint estimation of the parameters that index the mean and dispersion submodels. A novel feature of such residuals is they do not entail the computation of $n \times n$ projection matrices, which are computationally costly when n, the sample size, is large, as we shall later exemplify. Such matrices may require massive storage space when the number of data points in the sample is very large. In contrast, the residuals we propose only require the computation of $n \times 1$ vectors. Three empirical applications we present and discuss show that the proposed residuals successfully identify atypical and outlying observations. Interestingly, the evidence from such applications show that the two new residuals are complementary and should be used together. This was expected since each residual stems from Fisher's iterative algorithm for a different submodel (mean and dispersion). They thus carry different (and complementary) information on outlying observations and data patterns.

The remainder of the paper is organized as follows. Section 2 presents the class of nonlinear simplex regression models. In Section 3 we propose two new residuals for use with machine learning nonlinear simplex regressions. Empirical applications are presented and discussed. Finally, some concluding remarks and directions for future research are listed in Section 4.

2. The nonlinear simplex regression for machine learning

Let Y be a random variable with support in $(0, 1)$ and with density function

$$p(y;\mu,\sigma^2) = \{2\pi\sigma^2[y(1-y)]^3\}^{-1/2}\exp\left(-\frac{1}{2\sigma^2}d(y;\mu)\right), \quad 0 < y < 1, \qquad (1)$$

where $d(y;\mu)$, the deviance, is given by

$$d(y;\mu) = \frac{(y-\mu)^2}{y(1-y)\mu^2(1-\mu)^2}.$$

Using some results from Jørgensen (1997), it can be shown that $\mathbb{E}(Y) = \mu$ and

$$\mathrm{Var}(Y) = \mu(1-\mu) - \sqrt{\frac{1}{2\sigma^2}}\exp\left(\frac{1}{\sigma^2\mu^2(1-\mu)^2}\right)\Gamma\left(\frac{1}{2}, \frac{1}{2\sigma^2\mu^2(1-\mu)^2}\right),$$

where $\Gamma(a,b)$ is the incomplete gamma function defined as $\Gamma(a,b) = \int_b^\infty x^{a-1}e^{-x}dx$.

Let $y_1,\ldots,y_n$ be independent random variables such that y_t follows the density function given in (1) with parameters μ_t and σ_t^2. The nonlinear simplex regression for machine learning was proposed by Espinheira and Silva (2020) and is such that the mean and dispersion parameters satisfy

$$g(\mu_t) = f_1(x_t^\top;\beta) = \eta_{1t} \quad \text{and} \quad h(\sigma_t^2) = f_2(z_t^\top;\gamma) = \eta_{2t},$$

respectively, where $\beta = (\beta_1,\ldots,\beta_k)^\top \in \mathbb{R}^k$ and $\gamma = (\gamma_1,\ldots,\gamma_q)^\top \in \mathbb{R}^q$ are vectors of regression parameters ($k + q < n$). Additionally, η_{1t} and η_{2t} are nonlinear predictors, and $x_t = (x_{t1},\ldots,x_{tk_1})^\top$ and $z_t = (z_{t1},\ldots,z_{tq_1})^\top$ are observations on k_1 and q_1 known covariates ($k_1 \le k$ and $q_1 \le q$). The link functions $g : (0,1) \mapsto \mathbb{R}$ and $h : (0,\infty) \mapsto \mathbb{R}$ are strictly monotone and twice differentiable, and $f_i(\cdot)$, $i = 1, 2$, is continuous and differentiable such that the ranks of $J_1 = \partial\eta_1/\partial\beta$ and $J_2 = \partial\eta_2/\partial\gamma$ are k and q, respectively.

The parameters that index the model can be estimated by maximum likelihood. In the Appendix, we present the logarithm of the likelihood function, the score vector and Fisher's information matrix for the nonlinear simplex regression model.

3. New simplex residuals: proposal and applications

Model misspecification and atypical observations (anomalous cases) may impact both variability and bias of a machine learning process. The most direct way to validate the performance of a machine learning process applied to a particular dataset is by performing a residual analysis from which it should be possible to identify atypical observations and also to determine whether the fitted model at hand is misspecified, that is, to determine whether there are patterns in the data that should be incorporated to the model. The effectiveness of a residual analysis is heavily dependent on the residual in which it is based and its ability of mining data patterns and anomalous cases. Residual analyses are typically carried out by evaluating the agreement between the data and the fitted model, i.e., by comparing each response value to the corresponding fitted value (usually, the estimated mean). In the context of data mining, outlier maps based on descriptive tool plots constructed using robust residuals defined as $(y_t - \hat{\mu}_t)/\sqrt{\widehat{\mathrm{Var}}(y_t)}$ were proposed by Rousseeuw and van Zomeren (1990), where hats denote evaluation at parameter estimates.

In what follows we shall introduce two new residuals for the nonlinear simplex regression for machine learning. Espinheira and Silva (2020) considered the Fisher iterative mechanism for the estimation of β and obtained a residual from it which we shall denote by r_t^{β}. In what follows we shall consider a more general iterative scheme in which β and γ are jointly estimated. That is, unlike Espinheira and Silva (2020), we shall take into account information present in the estimation of dispersion effects when obtaining simplex residuals. By doing so, we shall obtain a new residual, namely: r_t^{γ}. This is the first new residual we introduce. The second new residual we propose follows from combining r_t^{β} and r_t^{γ} into a single residual.

Following Pregibon (1981), r_t^{β} and r_t^{γ} are derived from an artificial regression in which parameter estimation is performed by minimizing the sum of squared errors using Fisher's iterative scoring algorithm. Using Eq. (13) (see the Appendix), it can be shown that the mth iteration of Fisher's scoring algorithm for the estimation of β, the parameter vector in the nonlinear simplex mean submodel, is given by

$$\beta^{(m+1)} = \beta^{(m)} + (J_1^{(m)\top}\Sigma^{(m)}W^{(m)}J_1^{(m)})^{-1}J_1^{(m)\top}\Sigma^{(m)}T^{(m)}U^{(m)}(y-\mu^{(m)}), \tag{2}$$

where $J_1 = \partial\eta_1/\partial\beta$, $H = \mathrm{diag}\{1/g'(\mu_1),\ldots,1/g'(\mu_n)\}$, $\Sigma = \mathrm{diag}\{1/\sigma_1^2,\ldots,1/\sigma_n^2\}$, and U and W are diagonal matrices given in (11) and (12), respectively. The dimension of J_1 is $n\times k$.

Similarly, the mth iteration of Fisher's scoring algorithm for the estimation of γ, the parameter vector in the dispersion submodel, is

$$\gamma^{(m+1)} = \gamma^{(m)} + (J_2^{(m)\top}V^{(m)}J_2)^{-1}J_2^{(m)\top}H^{(m)}a^{(m)}, \tag{3}$$

where V is given in the Appendix, $H = \mathrm{diag}\{1/h'(\sigma_1^2),\ldots,1/h'(\sigma_n^2)\}$, $J_2 = \partial\eta_2/\partial\gamma$ is an $n\times q$ matrix of derivatives, and the tth element of a is

$$a_t = \frac{d(y_t;\mu_t)}{2(\sigma_t^2)^2} - \frac{1}{2\sigma_t^2}. \tag{4}$$

It is worth noticing that Eqs. (2) and (3) can be written in terms of weighted least squares estimators as $\beta^{(m+1)} = (J_1^\top\Sigma^{(m)}W^{(m)}J_1)^{-1}J_1^\top \Sigma^{(m)}W^{(m)}e_1^{(m)}$ and $\gamma^{(m+1)} = (J_2^\top V^{(m)}J_2)^{-1}J_2^\top V^{(m)}e_2^{(m)}$, respectively. Here, $e_1^{(m)} = J_1\beta^{(m)} + W^{-1^{(m)}}U^{(m)}T^{(m)}(y-\mu^{(m)})$ and $e_2^{(m)} = J_2\gamma^{(m)} + V^{-1^{(m)}}H^{(m)}a^{(m)}$.

Upon convergence, we obtain

$$\hat{\beta} = (\hat{J}_1^\top\hat{\Sigma}\hat{W}\hat{J}_1)^{-1}\hat{J}_1^\top\hat{\Sigma}\hat{W}e_1 \quad\text{and}\quad \hat{\gamma} = (\hat{J}_2^\top\hat{V}\hat{J}_2)^{-1}\hat{J}_2^\top\hat{V}e_2, \quad\text{where}\quad e_1 = \hat{\eta}_1 + \hat{W}^{-1}\hat{U}\hat{T}(y-\hat{\mu}) \quad\text{and}\quad e_2 = \hat{\eta}_2 + \hat{V}^{-1}\hat{H}\hat{a}. \tag{5}$$

Hereafter hats indicate evaluation at the maximum likelihood estimates.

Notice that $\hat{\beta}$ and $\hat{\gamma}$ in (5) are the least squares estimators of β and γ from the pseudo-regressions of $\hat{\Sigma}^{1/2}\hat{W}^{1/2}e_1$ on $\hat{\Sigma}^{1/2}\hat{W}^{1/2}J_1$ and of $\hat{V}^{1/2}e_2$ on $\hat{V}^{1/2}J_2$, respectively. By using regressions of pseudo-responses on pseudo-covariates it is possible to obtain closed-form expressions for the least squares residuals. The least squares residuals from the two separate regressions, i.e., from the regressions used to estimate β and γ, are $r^{\beta} = \hat{\Sigma}^{1/2}\hat{W}^{1/2}(e_1 - \hat{J}_1\hat{\beta}) = \hat{\Sigma}^{1/2}\hat{W}^{-1/2}\hat{T}\hat{U}(y-\hat{\mu})$ and $r^{\gamma} = \hat{V}^{1/2}(e_2 - \hat{J}_2\hat{\gamma}) = \hat{V}^{-1/2}\hat{H}\hat{a}$, respectively. Such residuals can be expressed as

$$r_t^{\beta} = \frac{y_t - \hat{\mu}_t}{\sqrt{\hat{v}_t/\hat{u}_t^2}} \tag{6}$$

and

$$r_t^{\gamma} = \frac{\hat{a}_t}{\sqrt{(2\hat{\sigma}_t^4)^{-1}}},$$

u_t and v_t being given in (11) and (12), respectively. Using the expression for a_t given in (4), it follows that

$$r_t^{\gamma} = \frac{d(y_t;\hat{\mu}_t) - \hat{\sigma}_t^2}{\hat{\sigma}_t^2\sqrt{2}}. \tag{7}$$

The residual r^{β} measures the degree of agreement between the tth response and the corresponding fitted mean value whereas r^{γ} measures the agreement between the tth deviance and the respective fitted dispersion. In what follows we shall refer to r^{β} and r^{γ} as 'bias residual' and 'variance residual', respectively.

Espinheira and Silva (2020) standardized r^{β} in (7) by the standard deviation of e_1. Notice that $\widehat{\mathrm{Cov}}(\hat{\beta}) \approx (J_1^\top\hat{\Sigma}\hat{W}J_1)^{-1}$, and hence $\widehat{\mathrm{Cov}}(e_1) \approx (\hat{\Sigma}\hat{W})^{-1}$. It then follows that $\widehat{\mathrm{Cov}}(r^{\beta}) = (1 - H_{\beta}^*)$, where $H_{\beta}^* = (\hat{W}\hat{\Sigma})^{1/2}\hat{J}_1(\hat{J}_1^\top\hat{\Sigma}\hat{W}\hat{J}_1)^{-1}\hat{J}_1^\top(\hat{\Sigma}\hat{W})^{1/2}$ is the projection matrix of the regression of $\hat{\Sigma}^{1/2}\hat{W}^{1/2}e_1$ on $\hat{\Sigma}^{1/2}\hat{W}^{1/2}\hat{J}_1$. The authors defined the following 'weighted residual':

$$r_{pt}^{\beta} = \frac{\hat{u}_t(y_t - \hat{\mu}_t)}{\sqrt{\hat{v}_t(1 - \hat{h}_{tt}^{\beta})}},$$

where $\hat{h}_{tt}^{\beta}$ is the tth diagonal element of H_{β}^*. We shall refer to r_{pt}^{β} as 'standardized bias residual'.

A clear disadvantage of the above residual is that it requires the computation of a projection matrix (H_{β}^*). As noted earlier, such computation can be very costly when the sample size is large. In order to illustrate the computation cost of storing projection matrices when the number of data point is large, consider, e.g., $n = 5000; 50{,}000; 500{,}000$. Since a double precision floating point number ('binary64') typically occupies 8 bytes of storage on modern computers, the projection matrices corresponding to these three sample sizes would require, respectively, 200 MB, 20 GB and 2 TB of computer memory. Prior to storing such a large matrix, it is, of course, necessary to compute it. Such computation requires: (i) multiplication of two n-diagonal matrices, (ii) computation of n square roots, (iii) multiplication of an $n\times k$ matrix by an n-diagonal matrix and then by an $n\times k$ matrix, (iv) computation of the inverse of the resulting $k\times k$ matrix, (v) multiplication of such a matrix by a $k\times n$ matrix, and then (vi) multiplication by the matrix obtained from (i) and (ii). It is noteworthy that it is common practice to construct normal probability plots with simulated envelopes for goodness-of-fit assessment. To that end, it is necessary to perform a large number of data resamples, and then estimate the model's parameters and compute the residuals for each pseudo-sample. The procedure becomes considerably more computationally expensive when it is necessary to compute a projection matrix for each generated sample.

We shall now introduce a new residual, the 'bias variance residual', by combining r_t^{β} and r_t^{γ}. To that end, we first consider $\hat{u}_t(y_t - \hat{\mu}_t) + \hat{a}_t$, and then we compute $\mathrm{Var}(\hat{u}_t(y_t - \hat{\mu}_t) + \hat{a}_t)$. Finally, the 'bias variance residual' is expressed as

$$r_t^{\beta\gamma} = \frac{\hat{u}_t(y_t - \hat{\mu}_t) + \hat{a}_t}{\sqrt{\hat{v}_t + (2\hat{\sigma}_t^4)^{-1}}}. \tag{8}$$

Note that the denominator involves $\widehat{\mathrm{Var}}(\hat{u}_t(y_t - \hat{\mu}_t)) + \widehat{\mathrm{Var}}(\hat{a}_t)$; we use the fact that $\widehat{\mathrm{Cov}}(\hat{u}_t(y_t - \hat{\mu}_t), \hat{a}_t) \approx 0$ which follows from β and γ being asymptotically orthogonal (Cox & Reid, 1987), as indicated in the Appendix. This is a very simple standardization when compared to that used in the combined residual (Espinheira et al., 2017) which is obtained in similar fashion.

It is noteworthy that it is possible to standardize the 'variance residual' using the projection matrix $H_{\gamma}^* = \hat{V}^{1/2}\hat{J}_2(\hat{J}_2^\top\hat{\Sigma}\hat{J}_2)^{-1}\hat{J}_2^\top\hat{V}^{1/2}$. By doing so we arrive at the 'standardized variance residual' defined as $r_{pt}^{\beta\gamma} = r_t^{\gamma}/\sqrt{1-\hat{h}_{tt}^{\gamma}}$. In what follows, we shall not use this residual because it requires the computation of a projection matrix. As noted earlier, such computation can be very costly when the sample size is large. As we shall see in the next section, even though the residual given in (8) does not incorporate a standardization term that includes information on the generalized leverage (matrices H_{β}^{*} and H_{γ}^{*}), it typically outperforms existing residuals when it comes to identifying cases that are influential.

Two other residuals that are already available in the literature are the deviance (Jørgensen, 1997) and quantile residuals (Dunn & Smyth, 1996) which are given, respectively, by

$$d(y_t,\hat{\mu}_t) = \frac{y_t-\hat{\mu}_t}{\hat{\mu}_t(1-\hat{\mu}_t)\sqrt{y_t(1-y_t)}} \quad \text{and} \quad r_t^{q} = \Phi^{-1}(\hat{p}_t),$$

where $\Phi(\cdot)$ and $p_t = F(y_t;\beta,\gamma)$ are the standard normal and simplex distribution functions.

We performed simulations to evaluate the mean, standard deviation, skewness and kurtosis of each residual (per observation). For brevity, we shall not present the results. Instead, we shall only outline the main conclusions obtained from such simulations. As expected, the quantile residual distribution is very well approximated by the standard normal distribution. The mean and standard deviation of the variance residual are close to zero and one, respectively. Such a residual displays, however, considerable positive skewness. All other residuals are also asymmetrically distributed. As a consequence, the thresholds that are commonly used for identifying atypical data points (i.e., −2 and 2) may not be accurate. We recommend that detection of atypical observations be based on the 0.025 and 0.975 empirical quantiles of the residuals, say $\omega_{(0.025)}$ and $\omega_{(0.975)}$, respectively, which are used in the construction of residual normal probability plots with simulated envelopes.

In what follows we shall present and discuss three empirical applications. The first one employs simulated data whereas the remaining two applications are based on observed data sets.

3.1. Simulated data – bias and high variance introduced by an outlier

We shall now use a simulated dataset to evaluate the new residuals' ability to identify outlying data points. We use the following varying dispersion simplex regression model:

$$\log\left(\frac{\mu_t}{1-\mu_t}\right) = \beta_1+\beta_2 x_{t2}+\beta_3 x_{t3} \quad \text{and} \quad \log(\sigma_t^2) = \gamma_1+\gamma_2 x_{t3},$$

$t = 1,\ldots,20$. The values of the covariates x_2 and x_3 were obtained as random draws, respectively, from the following distributions: exp(2) and $\mathcal{U}(0,1)$. The true parameter values are $\beta = (1.7, 1.1, 1.5)^{\top}$ and $\gamma = (-2.1, 4.1)^{\top}$, so that $y_t \in (0.7654, 0.9997)$, $y_{(0.5)} = 0.9146$ (median response) and $\lambda = \max\sigma_t^2/\min\sigma_t^2 = 52.04$. Fig. 1 contains the response (left panel) and covariates (right panel) boxplots.

The design of the simulated data introduces two relevant complexity conditions to the dataset. First, one of the values of x_3 is an outlier, namely: case 14, covariate value equals 4.22; the next largest value of x_3 is 1.34 (Fig. 1(b)). Second, the response variable values are close to the unit interval upper limit (Fig. 1(c)). Under such conditions the beta regression estimation process may display considerable numerical instability; see, e.g., Ospina et al. (2006).

Figs. 2 and 3 contain, respectively, index plots and normal probability plots with simulated envelopes ('envelope plots') constructed using the following residuals: bias, bias standardized, variance, bias variance, deviance and quantile. The deviance residual index plot singles out cases 10 and 12 as atypical data points. No other residual does so. All other residuals single out observation 1 as atypical, especially the variance and bias variance residuals. Recall that these two residuals do not use information from projection matrices. The variance residual index plot singles out cases $\{2, 14, 19\}$ as atypical in addition to observation 1.

Most data points are inside the normal probability plots with simulated envelopes (Fig. 3); few observations lie just outside the envelope bands. Such points are observations that have the potential to be influential. According to the variance residual probability plot (Fig. 3(c)), the following potentially influential data points are identified: $\{1, 6, 12, 17\}$. Case 1 is singled out in all plots whereas case 10 is only singled out in the bias standardized residual plot.

We removed the observations identified as atypical and possibly influential and estimated the model's parameters using the resulting incomplete data sets. Such data sets were obtained by jointly removing all atypical observations and also by removing all of their subsets; for brevity, we only present the scenarios for which the data points exclusion had the most noticeable impacts on inferential results. Table 1 contains the relative changes in the parameter estimates (%), the relative changes in the standard errors (%) and the p-values for the z tests that the relevant parameters equal zero. For reference, it also contains the parameter estimates, standard errors and p-values obtained using the complete data.

All parameters are statistically different from zero at the 1% significance level when estimation is carried using all observations. The parameters β_2 and γ_2 are overestimated, whereas γ_1 is underestimated. As noted earlier, the true ratio between the maximal and minimal values of σ_t^2 (λ) equals 52.04. This quantity is considerably overestimated, since the ratio between the largest and smallest values of $\hat{\sigma}_t^2$ (say, $\hat{\lambda}$) is 240.62.

When observation 1 is not in the data, the values of $\hat{\beta}_2$, $\hat{\gamma}_2$, $\hat{\gamma}_3$ increase by 11%, 45% and 39%, respectively. Additionally, the value of $\hat{\lambda}$ increases by over 750%. It is thus clear that observation 1 is influential. All parameters remain statistically different from zero at the 1% significance level. Removal of observation 14 causes a very large increase in the standard error of $\hat{\beta}_3$ (nearly 365%). The estimate of β_2 and the corresponding standard error are considerably impacted by the removal of observation 12. The same happens when data points 1 and 12 are jointly removed. Exclusion of case 10 does not impact inferential results noticeably; recall that this case was singled out as atypical by the deviance residual.

Clearly, observation 1 is the most influential data point. It is interesting to note that it was singled out as anomalous by all residuals but the deviance residual. Interestingly, the values of two proposed residuals for the first observation are much larger than those of the competing residuals. Hence, the new residuals provide more evidence of the anomaly of such an observation. As we have seen, case 14 has a very sizeable impact on the standard error of $\hat{\beta}_3$ (approximately 365%); such a case was only singled out as anomalous by one of the residuals we proposed (variance residual). We note that observation 14 corresponds to the largest values of x_3 and y.

This application showed that outliers may introduce both bias and high variability to simplex regression machine learning. The two new residuals emphatically highlighted the anomalous data points that impact these features of the model. It was shown, however, that the presence of such outliers in the data did not revert the hypothesis testing inferences used to validate the machine learning simplex process. That is, the z testing inferences are insensitive to extreme observations, and that can be taken as strong evidence that model predictions are not considerably impacted by atypical data points when the response distribution and the mean and dispersion predictors are correctly specified.

The most important aspect of this application is that it showcases the ability of the two new residuals (bias variance and variance) to mine anomalous cases that were not detected by alternative residuals. We note that, even though we based our analysis on visual inspection of residual plots, it is possible to perform intervention-free outlier detection by comparing each residual to the 0.025 ($\omega_{0.025}$) and 0.975 ($\omega_{0.975}$) quantiles of the empirical residual distribution which can be easily obtained by data resampling, as in the construction of envelope bands for normal probability plots.

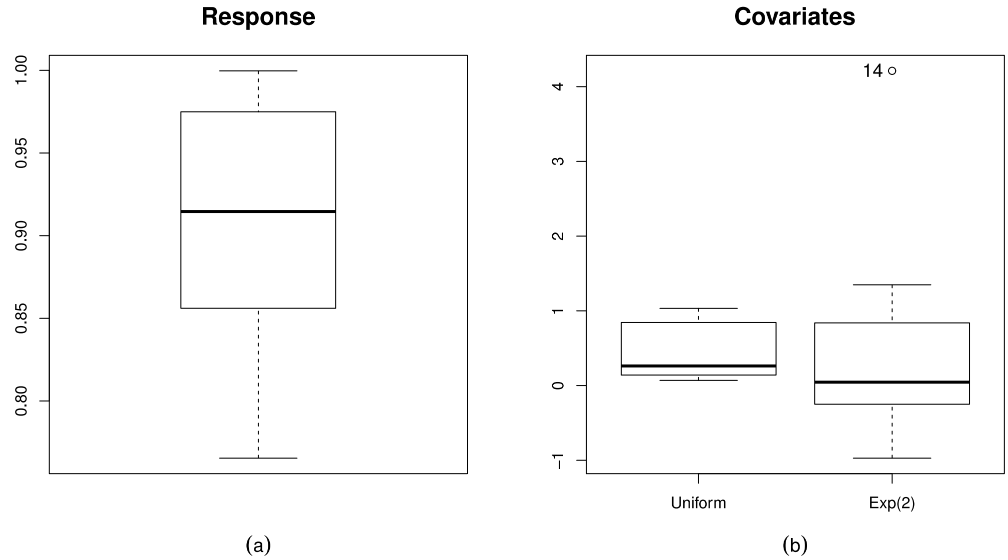

Fig. 1. Boxplots; simplex model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3}$ and $\log(\sigma_t^2) = \gamma_1 + \gamma_2 x_{t3}$, $t = 1, \ldots, 20$; simulated data.

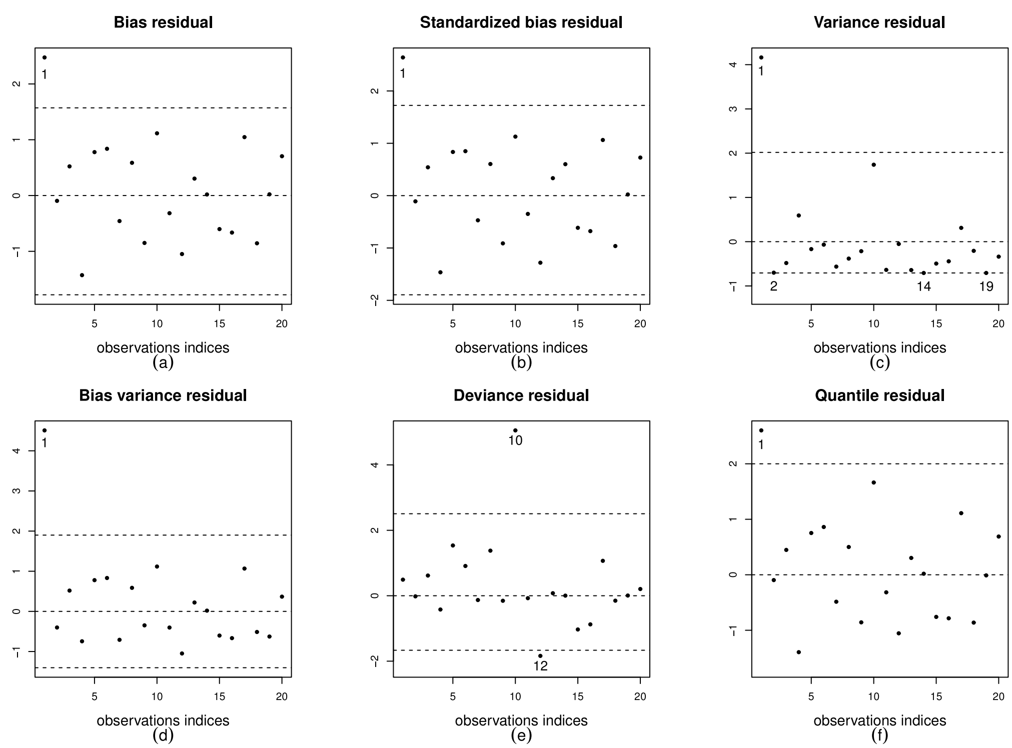

Fig. 2. Residual index plots; simplex model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3}$ and $\log(\sigma_t^2) = \gamma_1 + \gamma_2 x_{t3}$, $t = 1, \ldots, 20$; simulated data.

3.2. Oxidation of ammonia

We shall now model data on oxidation of ammonia collected during a period of 21 days of oxidation ($n = 21$) as a stage for nitric acid production in an industrial plant. The source of the data is Brownlee (1965). The variable of interest (response) is the proportion of ammonia that was not converted into nitric acid (y). It is a measure of the industrial plant total inefficiency. The covariates are: amount of normal atmospheric gas typically named as 'air current' (x_2), temperature of the water used for cooling the reaction (x_3), and acid concentration (x_4) measured as $10 \times$ (acid concentration $-$ 50). At the outset we consider the following linear simplex regression model with constant dispersion: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} + \beta_4 x_{t4} + \beta_5 (x_{t2} \times x_{t3})$. The parameter estimates (standard errors) [p-values] are $\hat{\beta}_1 = -12.001$ (1.884) [< 0.001], $\hat{\beta}_2 = 0.114$ (0.032) [< 0.001], $\hat{\beta}_3 = 0.229$ (0.090) [0.008], $\hat{\beta}_4 = -0.001$ (0.006) [0.867], $\hat{\beta}_5 = -0.003$ (0.001) [0.039] and $\hat{\sigma}^2 = 1.416$.

We note that acid concentration should be excluded from the mean submodel since the corresponding z test p-value is very large. However, before doing so it is useful to carry out a residual analysis of the fitted constant dispersion simplex model. In particular, we shall investigate whether there is evidence of nonconstant dispersion using residual normal probability plots with simulated envelopes. A crossing envelope pattern is indicative of incorrect dispersion specification. When that happens with a constant dispersion model, there is evidence of varying dispersion. This is exactly what happens in the residual normal probability plots presented in Fig. 4, except for the plot constructed using the

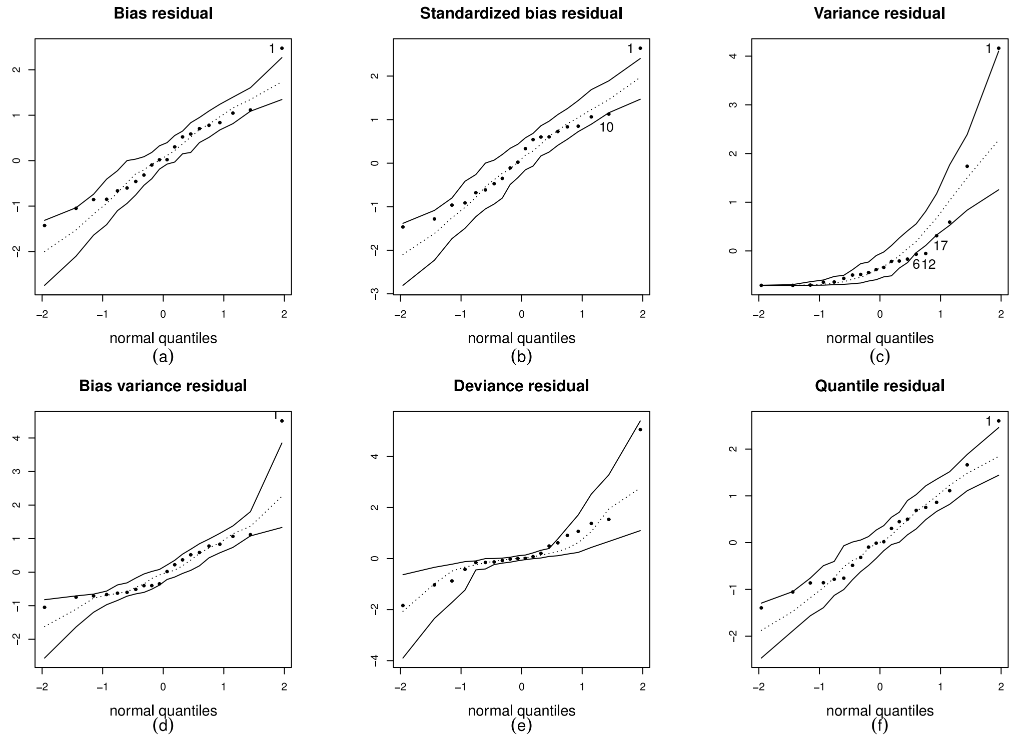

Fig. 3. Residual normal probability plots with simulated envelopes; simplex model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3}$ and $\log(\sigma_t^2) = \gamma_1 + \gamma_2 x_{t3}$, $t = 1, \ldots, 20$; simulated data.

Table 1
Parameter estimates, standard errors (s.e.), relative changes in the parameter estimates (%) and in the standard errors (%) due to cases exclusion and the respective p-values; simulated data.

$\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3}$ and $\log(\sigma_t^2) = \gamma_1 + \gamma_2 x_{t3}$						
Data	Quantity	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\gamma}_1$	$\hat{\gamma}_2$
Complete data	Estimate	1.673	1.233	1.508	−3.877	5.68
	s.e.	0.032	0.137	0.006	0.529	0.912
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001
obs. 1 deleted	est. change	−2.483	10.833	0.510	45.593	39.205
	s.e. change	−35.481	−22.562	−41.592	5.794	3.181
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001
obs. 10 deleted	est. change	0.302	−2.160	−0.050	−4.792	−14.716
	s.e. change	−4.719	−16.030	−0.557	1.557	7.908
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001
obs. 12 deleted	est. change	−1.495	11.736	0.236	−1.572	−3.673
	s.e. change	10.097	15.749	6.863	0.528	4.615
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001
obs. 14 deleted	est. change	0.660	0.947	−1.140	−2.815	−2.835
	s.e. change	16.702	0.363	364.545	6.273	3.619
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001
obs. 1 and 12 deleted	est. change	−3.321	17.673	0.639	44.408	34.522
	s.e. change	−31.999	−17.632	−39.195	6.368	7.818
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001

deviance residual. In particular, there is a very clear pattern of lower envelope crossing when the variance residual is used, i.e., in panel (c). Additionally, the residual of case 21 is way outside the envelope bands in such a plot which indicates that inferential conclusions drawn from the fitted model may be inaccurate. Overall, the variance residual is the best performing residual in the sense that it is the only residual that identifies an outlying data point and also because it most emphatically indicates the need of modeling the dispersion parameter. The atypicalness of case 21 may be attenuated by doing so. Recall that the variance residual was derived using information from Fisher's iterative scoring scheme applied to the dispersion submodel. It is then expected to be particularly sensitive to dispersion misspecification.

Since there is evidence of varying dispersion, we shall select covariates to be used in the dispersion submodel. To that end, we examined plots of the residuals against the explanatory variables which indicate that the data dispersion is inversely related to x_2. Nonetheless, such a pattern is even more evident when the residuals are plotted against the iteration between air current (x_2) and water temperature (x_3) when the variance and bias variance residuals are used; see Fig. 5. It is interesting to note that the small sample size makes it difficult to see that the dispersion of the data is considerably larger when the values of the interaction variable are smaller than 1600. However, even so, it can be concluded that the dispersion is not constant and that the interaction variable should be included in the dispersion submodel. It is noteworthy that case 21 is singled out as an outlier, especially in the variance residual plot, panel (a). We also note that case 4 is only singled out as atypical in the bias variance residual plot which shows, once again, that the two new residuals yield complementary information.

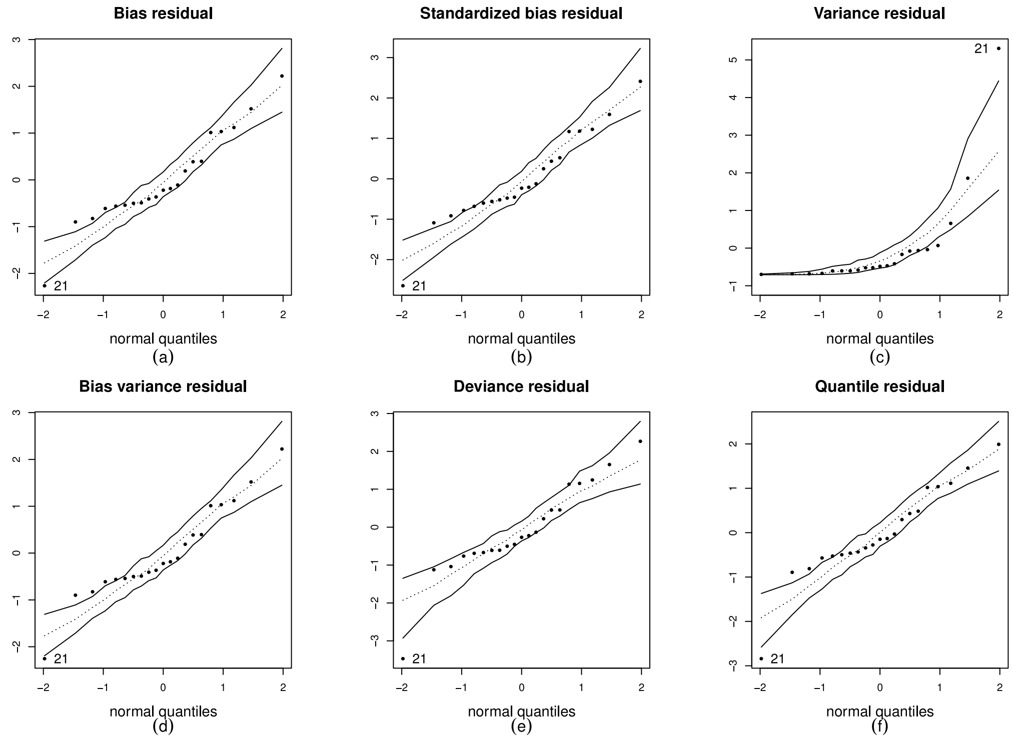

Fig. 4. Residual normal probability plots with simulated envelopes; simplex model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} + \beta_4(x_{t2} \times x_{t3})$, constant dispersion, $t = 1, \ldots, 21$; data on oxidation of ammonia.

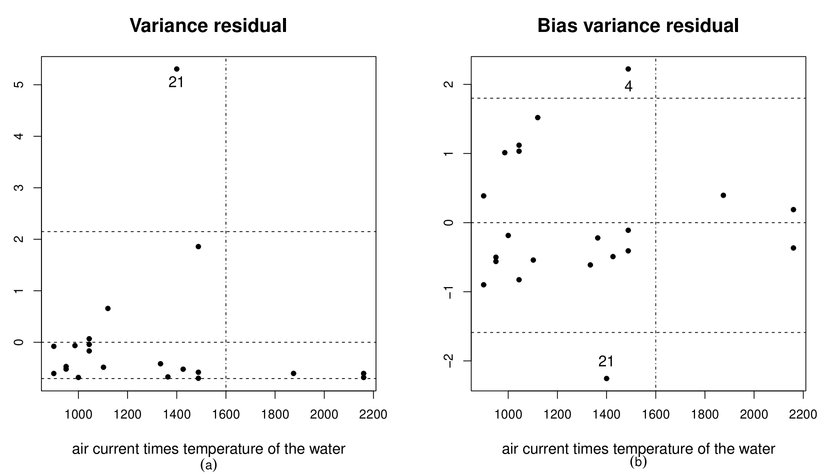

Fig. 5. Residual versus iteration between air current and temperature of the water; simplex model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} + \beta_4(x_{t2} \times x_{t3})$, constant dispersion, $t = 1, \ldots, 21$; data on oxidation of ammonia.

On the basis of the residual analysis described above, we arrived at the following varying dispersion simplex regression model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} + \beta_4 x_{t4} + \beta_5(x_{t2} \times x_{t3})$ and $\log(\sigma_t^2) = \gamma_1 + \gamma_2 x_{t3} + \gamma_3(x_{t2} \times x_{t3})$. Since the z tests p-values for testing that β_5 and γ_1 equal zero are considerably larger than 0.10, we decided to proceed with the following model:

$$\log\left(\frac{\mu_t}{1-\mu_t}\right) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} + \beta_4(x_{t2} \times x_{t3}),$$
$$\log(\sigma_t^2) = \gamma_1 x_{t3} + \gamma_2(x_{t2} \times x_{t3}). \tag{9}$$

The model parameters were estimated by maximum likelihood and the estimated degree of nonconstant dispersion is $\hat{\lambda} = 16.214$, with $\hat{\sigma}^2_{\min} = 0.536$ and $\hat{\sigma}^2_{\max} = 8.686$. In order to evaluate the model fit, we plotted the residuals against the observations indices; see Fig. 6. The residuals appear to be randomly scattered around zero, the following observations showing up as atypical: $\{1, 2, 4, 19, 20, 21\}$. It is noteworthy that the variance residual provides more evidence of the anomaly of data point 21 than the competing residuals. The same residual indicates that cases 1, 2 and 19 are also atypical. Additionally, the only atypical observation according to the quantile residual is case 21.

Fig. 7 contains normal probability plots with simulated envelopes. All data points but case 3 lie inside the envelopes for the bias, bias standardized and bias variance residuals; observation 3 lies slightly below the lower envelope bands of such plots. Additionally, when the residual plot is constructed using the variance residual, observations $\{1, 2, 3, 8, 19\}$ are very close to the lower envelope band. They may be influential and are worthy of further examination. The quantile residual probability plot indicates that only case 21 is atypical. No data point is atypical according to the deviance residual. It worth noticing that case 21 is not singled out as anomalous in Fig. 7. Even though it is atypical, its inferential impact is attenuated by the dispersion modeling, especially its impact on the mean parameter estimates.

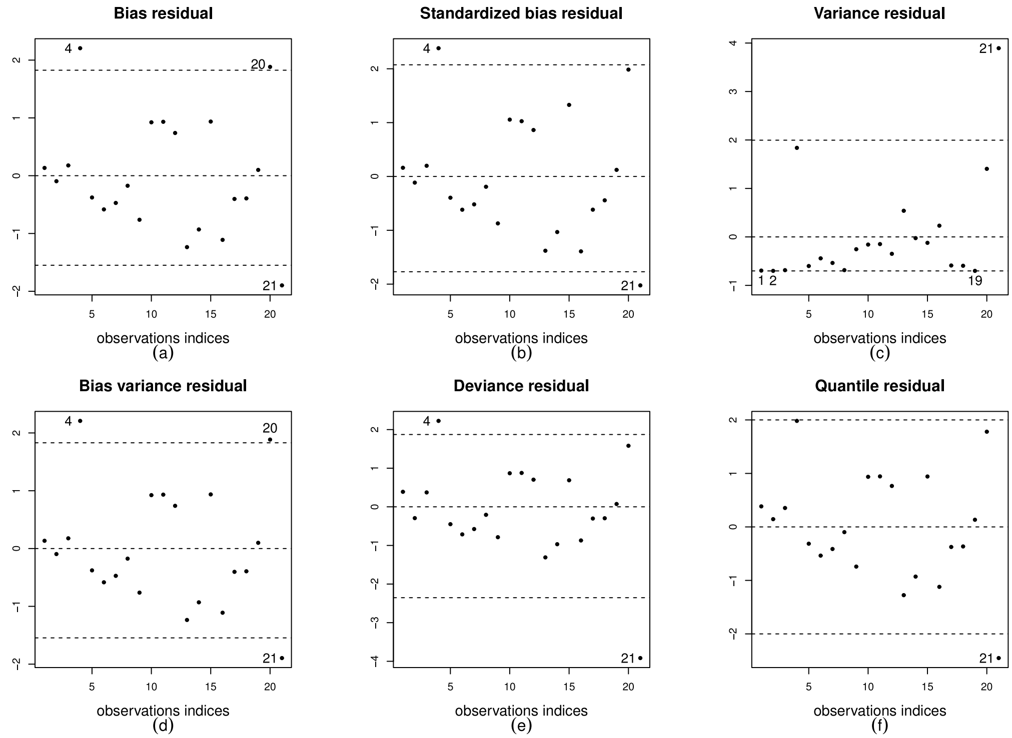

Fig. 6. Residual index plots; simplex model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{2t} + \beta_3 x_{3t} + \beta_4(x_{2t} \times x_{3t})$ and $\log(\sigma_t^2) = \gamma_2 x_{t3} + \gamma_2(x_{2t} \times x_{3t})$, $t = 1, \ldots, 21$; data on oxidation of ammonia.

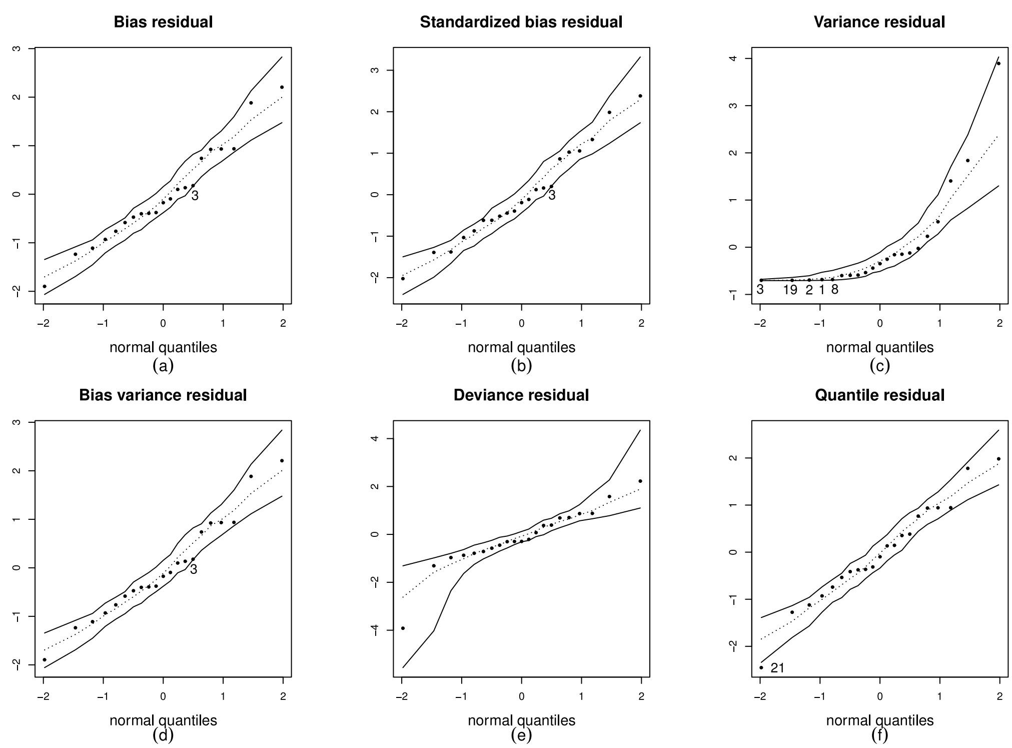

Fig. 7. Residual normal probability plots with simulated envelopes; simplex model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} + \beta_4(x_{t2} \times x_{t3})$ and $\log(\sigma_t^2) = \gamma_2 x_{t3} + \gamma_3(x_{t2} \times x_{t3})$, $t = 1, \ldots, 21$; data on oxidation of ammonia.

In order to determine whether the observations identified as atypical are influential, we removed them from the data and estimated the model parameters using the resulting incomplete datasets. Table 2 contains the relative changes in the point estimates and in the corresponding standard errors (%) and also the z tests p-values. It is noteworthy that some point estimates and p-values change considerably when observations 1, 3, 20 and/or 21 are not in the data. For instance, when observations 1 and 3 are individually removed from the data the p-values of the test that β_4 equals zero jump from 0.050 to 0.086 and 0.075, respectively.

The individual removal of observations 20 and 21 also impact the inference, especially the removal of case 20. Data point 21 exerts substantial influence on the dispersion (but not on the mean) parameter estimates. In contrast, observation 20 is influential for the mean submodel estimates. When it is removed from the data, the interaction covariate p-value in the mean submodel jumps to 0.119. As a consequence, the null hypothesis is no longer rejected at the 10% nominal level. We note that the z test is based on a large sample approximation and may not be accurate in small samples. In that sense, statistical significance should be assessed with care, and p-values close to 0.10 are indicative of plausible significance.

The joint exclusion of cases 4 and 20 leads to substantial changes in the standard errors of the estimated coefficients in the mean submodel. Cases 1 and 2 are jointly even more harmful, since their joint exclusion causes large changes in all parameter estimates but the intercept and also in all standard errors; additionally, the p-value of the test that β_4 equals zero jumps from 0.05 to 0.10.

The variance residual thus indicate that observations 1 and 2 are atypical, and they proved to also be influential. According to the bias, bias standardized and bias variance residuals, case 3 is atypical and it proved to be influential. It is thus clear that diagnostic analyses based on the variance and bias variance residuals outperform those based on the deviance and quantile residuals.

We have also fitted a beta regression model to the data. Initially, we used the same model specification as that of the simplex model. The p-value for the test that β_4 equals zero is 0.140 when all observations are used. When observation 10 (20) is not in the data such a p-value becomes 0.153 (0.234). When the two observations are jointly removed from the data, the p-value increases to 0.300. It is thus clear that the inferential impact of the two influential observations is quite pronounced in the beta model.

We considered alternative beta regression formulations and arrived at the following model, for which all covariates are statistically significant at the 5% significance level:

$$\log\left(\frac{\mu_t}{1-\mu_t}\right) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} \quad \text{and} \quad \log(\phi_t) = \gamma_1 + \gamma_2 x_{t2}, \qquad (10)$$

where μ_t and ϕ_t are the beta distribution mean and precision parameters for the tth response.

Fig. 8 presents the normal probability plot with simulated envelopes using the quantile residual for Model (10). We note that several points lie outside the simulated envelope which is evidence of poor model fit. Similar patterns were found for other beta regression models. Since different regressors were considered for use in the mean and precision submodels and in all configurations several residuals fell outside the envelope bands, there is evidence that the beta law is not adequate for fitting the data at hand.

We computed the ratio between the maximal and minimal estimated response variances. For the beta model (Model (10)), it is approximately equal to 5200 whereas for the simplex model (Model (9)) it is close to 5. This ratio is known as the degree of heterogeneity. The simplex model thus displays substantially less heterogeneity than the beta model. It is noteworthy that all response values are close to the standard unit interval lower limit since $\min(y_t) = 0.007$ and $\max(y_t) = 0.042$, and high variability at the extremes of the unit interval is not expected. The substantially smaller heterogeneity intensity of the simplex model relative to beta model may be taken as evidence in favor of the former.

Espinheira and Silva (2020) showed that when all response values are close to one and there are influential data points, the maximum likelihood estimates of the simplex model parameters tend to be more stable (i.e., less influenced by atypical observations) than those of the beta model. The above application showed that such a stability gain also takes place when the response values are close to zero.

3.3. Vanadium data

The FCC (Fluid Catalytic Cracking) process or catalytic rupture is used to convert high molecular weight hydrocarbons into small molecules of higher commercial value by exposing them to a catalyst. In particular, it is used to convert heavy oil fractions into gasoline, C3 and C4 olefins, LPG and other ingredients that allow the formulation of diesel fuel. The FCC process is often considered to be the central pillar of a refinery, since it is associated with the generation of products in higher demand and/or more profitable (Garcia Salazar, 2005). One of catalyst main components is zeolite Y.

Another chemical component used in the FCC process is vanadium. It is well known that the addition of 1000 ppm of vanadium to the catalyst reduces the amount of gasoline produced by approximately 2.3%. Additionally, vanadium is known to contribute to the destruction of the catalyst, thus reducing the active surface and the selectivity and crystallinity of zeolite Y, especially in the presence of water vapor. Such a reaction also depends on the temperature which should be close to 720 degrees Celsius (Garcia Salazar, 2005).

Our interest lies in modeling the proportion of zeolite Y, which is the response variable (y). The following covariates are used: amount of water vapor (x_2), temperature (x_3) and vanadium concentration (x_4). There are 28 observations. Espinheira and Silva (2020) modeled such data. At the outset, they considered the following linear model:

$$\log\left(\frac{\mu_t}{1-\mu_t}\right) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} + \beta_4 x_{t4},$$
$$\log(\sigma_t^2) = \gamma_1 + \gamma_2 x_{t4}.$$

They then noticed that the residual index plot displayed a pattern that is typical of neglected nonlinearity. We shall use the same residual as the authors and also the two residuals proposed in this paper to determine whether the true functional form is linear or nonlinear; if the latter, we shall seek to determine which regression coefficient induces the nonlinear behavior.

Fig. 9(a) is the same as presented in Espinheira and Silva (2020). It shows a clear nonlinear pattern in the index plot constructed using their bias residual. Nonetheless, such a pattern is even more evident in the bias variance residual index plot; see Fig. 9(c). In contrast, the variance residual index plot shows no detectable pattern; Fig. 9(b). The information uncovered by the joint examination of the two index plots of the residuals we propose is extremely useful from a modeling viewpoint, namely: it is possible to conclude that there is nonlinearity only in the mean predictor, not in the dispersion predictor. We shall then proceed by using such a piece of information.

Next, we searched for a nonlinear simplex model that would provide a good data fit. As explained above, we restrict attention to models in which only the mean prediction is nonlinear; the dispersion predictor is taken to be linear. We considered as baseline the model used by Espinheira and Silva (2020) and arrived at the following nonlinear model:

$$\log\left(\frac{\mu_t}{1-\mu_t}\right) = \beta_1 + \beta_2 x_{t2}/(x_{t2} + \beta_3) + \beta_4 x_{t3} + \beta_5 \sqrt{x_{t4}},$$
$$\log(\sigma_t^2) = \gamma_1 + \gamma_2 x_{t4}^2.$$

In order to perform parameter estimation, we first need to obtain starting values for the model's parameters. The tth row of $J_1^{(0)} = [\partial\eta/\partial\beta]_{\beta=\beta^{(0)}}$ is $\left(1, x_{t2}/\left(x_{t2}+\beta_2^{(0)}\right), -\beta_1^{(0)} x_{t2}/\left(x_{t2}+\beta_2^{(0)}\right)^{-2}, x_{t3}, \sqrt{x_{t4}}\right)$. Since there are more parameters than covariates, it is possible to obtain $\left(\beta_1^{(0)}, \beta_2^{(0)}, \beta_4^{(0)}, \beta_5^{(0)}\right)^\top$ as $(X^\top X)^{-1} X^\top g(y)$, where tth row of X is $x_t^\top = \left(1, x_{t2}/\left(x_{t2} + \beta_3^{(0)}\right), x_{t3}, \sqrt{x_{t4}}\right)$, provided we set a value for $\beta_3^{(0)}$. We note that the water vapor covariate assumes values in $(-20, 80)$ and also that the following restrictions hold: $\left(x_{t2} + \beta_3^{(0)}\right) = -20$ and $\left(x_{t2} + \beta_3^{(0)}\right) = 80$. When the value of x_{t2} is minimal, $\beta_3^{(0)} \in (-20.80)$; when such a value is maximal, $\beta_3^{(0)} \in (-75.8.24.2)$. We experimented with different values for $\beta_3^{(0)}$ in the interval $(-20, 24.2)$, and $\beta_3^{(0)} = -19$ yielded

Table 2
Parameter estimates, standard errors (s.e.), relative changes in the parameter estimates (%) and in the standard errors (%) due to cases exclusion and the respective p-values; data on oxidation of ammonia.

$\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} + \beta_4(x_{t2} \times x_{t3})$ and $\log(\sigma_t^2) = \gamma_1 x_{t3} + \gamma_2(x_{t2} \times x_{t3})$							
Data	Quantity	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_4$	$\hat{\gamma}_1$	$\hat{\gamma}_2$
Complete data	Estimate	−13.272	0.137	0.283	−0.004	−0.217	0.004
	s.e.	2.401	0.041	0.121	0.002	0.094	0.002
	p-value	0.000	0.001	0.019	0.050	0.021	0.013
obs. 1 deleted	est. change	4.627	8.225	9.718	12.685	47.743	47.049
	s.e. change	24.588	26.206	26.663	29.161	13.768	16.683
	p-value	0.000	0.003	0.042	0.086	0.002	0.001
obs. 3 deleted	est. change	3.592	6.346	7.699	9.987	31.856	31.633
	s.e. change	17.978	19.019	19.751	21.470	4.755	6.268
	p-value	0.000	0.003	0.035	0.075	0.004	0.002
obs. 4 deleted	est. change	−3.986	−4.357	−8.702	−6.297	−3.019	−7.720
	s.e. change	−15.320	−15.523	−15.926	−16.348	0.142	0.011
	p-value	0.000	0.000	0.011	0.028	0.026	0.021
obs. 20 deleted	est. change	−7.021	−11.289	−16.463	−19.140	22.204	18.330
	s.e. change	0.845	1.288	1.305	1.986	1.487	1.008
	p-value	0.000	0.003	0.053	0.119	0.006	0.004
obs. 21 deleted	est. change	3.691	6.981	5.759	8.117	−157.858	−161.432
	s.e. change	−16.143	−41.913	−50.771	−52.207	0.809	1.304
	p-value	0.000	0.000	0.000	0.000	0.187	0.130
obs. {4, 20} deleted	est. change	−9.099	−12.484	−20.385	−19.547	25.929	14.015
	s.e. change	−20.613	−20.306	−20.759	−20.432	1.683	1.033
	p-value	0.000	0.000	0.019	0.047	0.004	0.005
obs. {1, 2} deleted	est. change	7.421	13.325	15.374	20.281	99.215	98.820
	s.e. change	35.838	39.487	38.868	43.886	45.578	54.103
	p-value	0.000	0.006	0.051	0.100	0.001	0.001

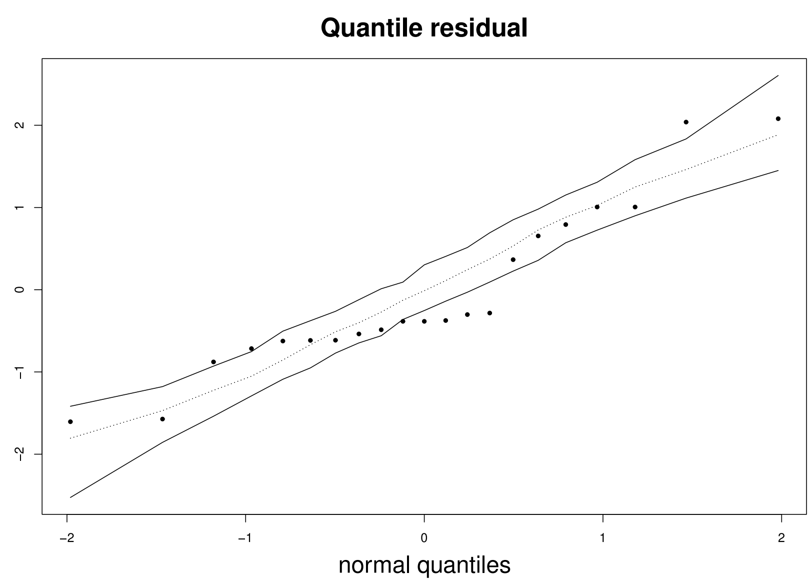

Fig. 8. Residual normal probability plot with simulated envelopes; beta model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3}$ and $\log(\phi_t) = \gamma_1 + \gamma_2 x_{t2}$, $t = 1, \ldots, 21$; data on oxidation of ammonia.

convergence of the iterative estimation mechanism and also plausible parameter estimates and standard errors. After selecting a value for $\beta_3^{(0)}$, we computed $\beta_{NL}^{(0)} = \left(J_1^{(0)\top} J_1^{(0)}\right)^{-1} J_1^{(0)\top}\left(g(y) - f\left(x, \beta_L^{(0)}\right)\right)$, where $f\left(x, \beta_L^{(0)}\right) = \beta_1^{(0)} + \beta_2^{(0)} x_{t2}/\left(x_{t2} + \beta_3^{(0)}\right) + \beta_4^{(0)} x_{t3} + \beta_5^{(0)}\sqrt{x_{t4}}$.

Upon convergence of the maximum likelihood iterative estimation scheme, we constructed residual index plots; see Fig. 10(a), (b) and (c). The residuals appear to be randomly scattered around zero. The bias and bias variance residuals identify observation 10 as atypical whereas the variance residual quite clearly identifies case 1 as atypical, and less strongly cases 4, 8, 14 and 22. The normal probability plots with simulated envelopes constructed using the two residuals are presented in Fig. 10(d), (e) and (f). Four observations lie outside the simulated envelope in the bias variance residual normal probability plot, namely: cases 1, 13, 14 and 27. They may be influential and are worthy of further examination.

We sequentially excluded the highlighted cases from the data, estimated the model's parameters again and computed the relative changes (%) in the point estimates. There was no substantial change in the mean submodel point estimates or p-values which is evidence that none of the identified observations is individually influential.

We then investigated whether some of the four observations identified as atypical or even all four data points are jointly influential. To that end, we sequentially removed all pairs and triplets of the highlighted observations and also all four data points simultaneously. For each configuration, we computed the relative changes in the parameter estimates and standard errors (%) and the corresponding p-values. The results are presented in Table 3. The inferences are greatly impacted by

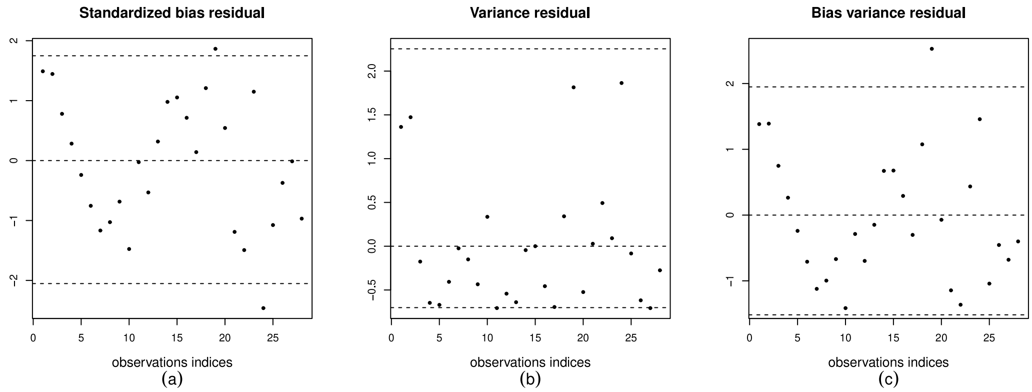

Fig. 9. Residual index plots; linear simplex model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} + \beta_4 x_{t4}$ and $\log(\sigma_t^2) = \gamma_1 + \gamma_2 x_{t4}$, $t = 1, \ldots, 28$; FCC data.

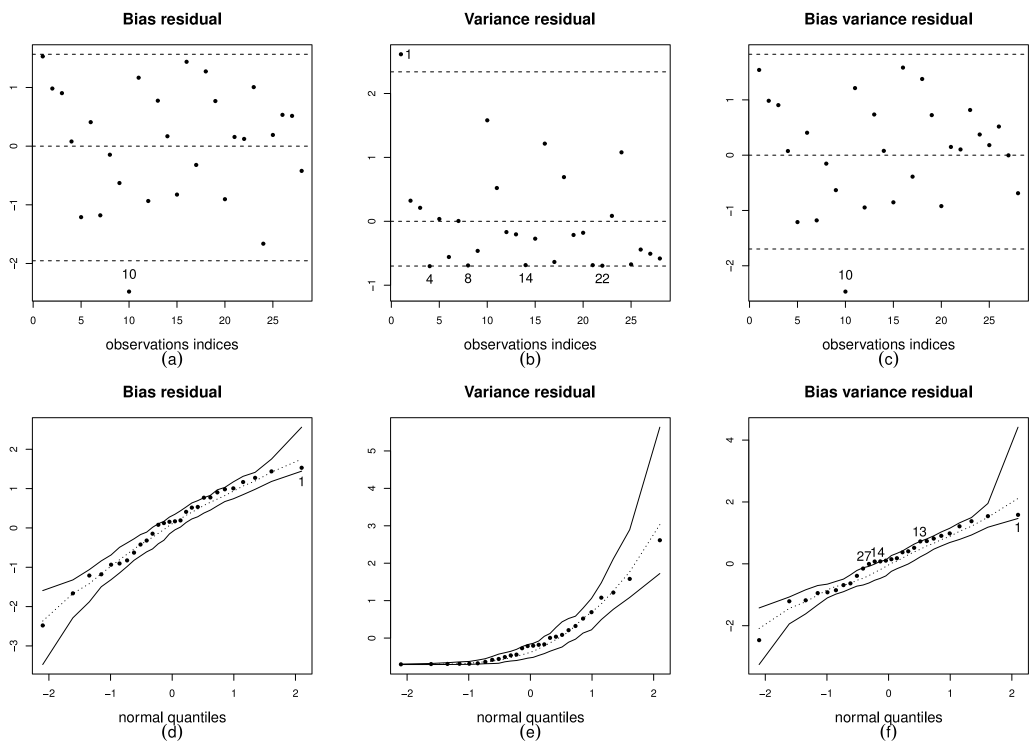

Fig. 10. Residual index and normal probability plots; nonlinear simplex model: $\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2}/(x_{t2} + \beta_3) + \beta_4 x_{t3} + \beta_5\sqrt{x_{t4}}$ and $\log(\sigma_t^2) = \gamma_1 + \gamma_2 x_{t4}^2$, $t = 1, \ldots, 28$; FCC data.

the removal of all four data points (i.e., observations $\{10, 13, 14, 27\}$). For instance, the p-value associated with the water vapor covariate jumps from 0.012 to 0.034 and the p-value relative to the vanadium concentration regressor goes from 0.007 to 0.021.

The new residuals were thus more capable of identifying influential data points than the bias residual. It is also worth noticing that no testing inference carried out at the 5% significance level was reversed which indicates that the nonlinear simplex model at hand provides a good fit to the data. Espinheira and Silva (2020) performed a local influence analysis for that model and also for a competing nonlinear beta regression model and found the simplex parameter estimation process to be more stable than that of the beta model when the data contain influential cases. Finally, we note that all response values belong to the upper half of the standard interval, with $\min(y_t) = 0.643$ and $\max(y_t) = 0.963$. Also, $\hat{\sigma}^2_{\min} = 0.088$ and $\hat{\sigma}^2_{\max} = 2.281$. When the values of σ^2 are in this range, the simplex distribution typically yields very good fits; see, e.g., Zhang et al. (2016).

It is noteworthy that the variance residual was the only residual that, based on the normal probability plot with simulated envelopes, identified cases 13, 14 and 27 as atypical, and such points proved to be quite influential. In particular, these cases considerably impacted the model variability and the tests p-values. It is also important to highlight that the simplex learning process was able to adequately accommodate the outlying observations in the sense that such cases did not alter hypothesis testing inferences made on the model's parameters even though two extreme complexity features were at play, namely: the response values are close to one and there is nonlinearity in the mean submodel.

4. Concluding remarks and directions for future research

We proposed two new residuals, the variance residual and the bias variance residual, for use with nonlinear simplex regression models. The new residuals were obtained from the iterative scoring scheme used for the joint estimation of β and γ, the parameter vectors that index the mean and dispersion submodels. Espinheira and Silva (2020) obtained simplex regression residuals using the iterative score algorithm for β (the bias and standardized bias residuals). In contrast, one of the

Table 3
Parameter estimates, standard errors (s.e.), relative changes in the parameter estimates (%) and in the standard errors (%) due to cases exclusion and the respective p-values; FCC data.

$\log(\mu_t/(1-\mu_t)) = \beta_1 + \beta_2 x_{t2}/(x_{t2}+\beta_3) + \beta_4 x_{t3} + \beta_5\sqrt{x_{t4}}$ and $\log(\sigma_t^2) = \gamma_1 + \gamma_1 x_{t4}^2$

Data	Quantity	$\hat\beta_1$	$\hat\beta_2$	$\hat\beta_3$	$\hat\beta_4$	$\hat\beta_5$	$\hat\gamma_1$	$\hat\gamma_2$
Complete data	Estimate	2.375	−0.107	−27.803	−0.290	−0.751	0.825	−1.217
	s.e.	0.149	0.042	4.021	0.107	0.142	0.370	0.314
	p-value	0.000	0.012	0.000	0.007	0.000	0.026	0.000
obs. {1} deleted	est. change	−4.415	0.279	−3.852	3.177	−10.307	−55.646	−26.530
	s.e. change	−0.893	−20.772	2.015	2.655	−10.321	3.604	1.737
	p-value	0.000	0.001	0.000	0.006	0.000	0.339	0.005
obs. {14, 27} deleted	est. change	0.638	2.680	−0.495	10.955	0.689	−1.218	−7.497
	s.e. change	5.219	4.068	7.810	14.229	2.143	0.484	5.362
	p-value	0.000	0.013	0.000	0.008	0.000	0.028	0.001
obs. {13, 14, 27} deleted	est. change	0.163	4.123	0.295	3.494	0.205	3.991	−4.090
	s.e. change	6.295	7.861	5.076	14.839	3.719	0.754	6.063
	p-value	0.000	0.015	0.000	0.014	0.000	0.021	0.001
obs. {10, 13, 27} deleted	est. change	3.805	−6.831	3.258	−5.983	10.781	−17.550	−6.400
	s.e. change	−4.766	6.955	−4.940	5.886	−4.365	4.144	7.078
	p-value	0.000	0.028	0.000	0.016	0.000	0.078	0.001
obs. {10, 14, 27} deleted	est. change	4.002	−7.065	2.051	0.543	10.945	−20.032	−11.680
	s.e. change	−2.562	8.803	6.420	10.115	−3.597	4.144	7.078
	p-value	0.000	0.031	0.000	0.013	0.000	0.087	0.001
obs. {10, 13, 14, 27} deleted	est. change	3.662	−5.959	2.780	−5.961	10.818	−15.409	−8.776
	s.e. change	−1.542	11.840	3.504	11.033	−2.286	4.445	7.726
	p-value	0.000	0.034	0.000	0.021	0.000	0.071	0.001

residuals we proposed (the bias variance residual) uses information from both iterative score algorithms, i.e., the new residual captures sample evidence from the joint estimation of two distribution parameters (mean and dispersion), and not only from the estimation of the location parameter.

We presented and discussed three empirical applications, one based on simulated data and two based on observed data. In such applications, we used the two residuals we proposed and also four residuals that are already available in the literature for use with simplex regression models, namely: the bias, standardized bias, deviance and quantile residuals. The new residuals performed quite well in all three applications. That is, the variance residual and the bias variance residual proved to be very effective for: (i) detecting anomalous cases and (ii) mining data patterns not accounted for. In particular, pattern detection indicated the need to model the data dispersion which in turn led to inferential results that were stable even in the presence of outliers. Additionally, a clear nonlinear pattern in the bias variance residual index plot indicated that the mean predictor should account for existing nonlinearity. It is noteworthy that nonlinearity is a high complexity factor and that one of the residuals proposed in this paper was able to uncover it by mining the data. It is also important to note that our empirical evidence indicates that the two new residuals are complementary and should be used together in residual analyses, since they identified different cases which then proved to be individually and jointly influential for the simplex mean and dispersion learning processes. We strongly recommend that diagnostic analyses carried out for simplex learning processes be based on the two new residuals.

Parameter estimation for the simplex regression model proved to be reliable and stable when the response values are close to the one of the standard unit interval limits which is an advantage of the simplex model relative to the beta model. Interestingly, in an empirical application in which the response values are close to zero, the ratio between the maximal and minimal estimated response variances was much smaller for the simplex model than for the beta model (approximately 5 vs. approximately 5200 in one of the applications). The residual normal probability plot with simulated envelopes indicated that the beta law was not suited for the data at hand and the z test p-values proved to be unstable in face of outlying cases. We note that z tests tend to be somewhat insensitive to outlying cases provided that the postulated model yields a very close approximation to the true data generating process. To that end, it is required that the underlying probability distribution, the mean predictor and the precision/dispersion predictor be correctly specified, and that is the chief goal of model building guided by residual analyses.

There are some directions that can be explored in future research. First, it would be useful to develop misspecification tests for varying dispersion nonlinear simplex regressions. Such tests can be used to determine whether the model specification is in error within the framework of a hypothesis test. Second, it would be useful to define simplex models for non-independent random variables, in particular, models that can be used to perform time series analysis and forecasting. A dynamic beta model was introduced by Rocha and Cribari-Neto (2009, 2017). Third, we plan to use of the 'prediction coefficient' (P^2) for selecting the simplex model with highest predictive accuracy, as did Espinheira et al. (2019) in the context of beta regression for machine learning. Fourth, we plan to construct prediction intervals for simplex regressions and numerically evaluate their accuracy. The main goal is to predict missing response values for which values on the covariates are available, as in Espinheira et al. (2014). Finally, we shall consider boosted simplex regression along the lines of Schmid et al. (2013) so that estimation and variable selection can be carried out efficiently and simultaneously.

CRediT authorship contribution statement

Patrícia L. Espinheira: Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Luana C.M. Silva:** Formal analysis, Investigation, Methodology, Visualization, Writing - original draft. **Francisco Cribari-Neto:** Formal analysis, Investigation, Methodology, Supervision, Visualization, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We gratefully acknowledge financial support from Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq). We also thank two anonymous referees for comments and suggestions that led to a much improved manuscript.

Appendix. Fisher's iterative scoring algorithm

In what follows we shall present the score function and Fisher's information for β and γ for the nonlinear simplex regression model (Espinheira & Silva, 2020). The log-likelihood function for Model (1) based on a sample of n independent responses is $\ell(\beta,\gamma) = \sum_{t=1}^{n} \ell_t(\mu_t,\sigma_t^2)$, where $\ell_t(\mu_t,\sigma_t^2) = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log(\sigma_t^2) - \frac{3}{2}\log[y_t(1-y_t)] - \frac{1}{2\sigma_t^2} d(y_t;\mu_t)$. The score function for β is $U_\beta(\beta,\gamma) = J_1^\top \Sigma T U(y-\mu)$, where $T = \mathrm{diag}\{1/g'(\mu_1),\ldots,1/g'(\mu_n)\}$ and

$$U = \mathrm{diag}\{u_1,\ldots,u_n\} \quad \text{with} \quad u_t = \frac{1}{\mu_t(1-\mu_t)}\left[d(y_t;\mu_t) + \frac{1}{\mu_t^2(1-\mu_t)^2}\right]. \tag{11}$$

The score function for γ can be written as $U_\gamma(\beta,\gamma) = J_2^\top H a$, where a_t is given in (4). The relevant blocks of Fisher's information matrix are $K_{\beta\beta} = J_1^\top \Sigma W J_1$ $K_{\beta\gamma} = K_{\gamma\beta}^\top = 0$ and $K_{\gamma\gamma} = J_2^\top V J_2$, where

$$W = \mathrm{diag}\{w_1,\ldots,w_n\} \quad \text{with} \quad w_t = \frac{v_t}{\sigma_t^2[g'(\mu_t)]^2},$$

$$v_t = \sigma_t^2\left[\frac{3\sigma_t^2}{\mu_t(1-\mu_t)} + \frac{1}{\mu_t^3(1-\mu_t)^3}\right]$$

$$\text{and} \quad V = \mathrm{diag}\{v_1,\ldots,v_n\} \quad \text{with} \quad v_t = \frac{1}{2(\sigma_t^2)^2}\frac{1}{[h'(\sigma_t^2)]^2}. \tag{12}$$

The iterative scoring schemes used for estimating β and γ can be written, respectively, as

$$\beta^{(m+1)} = \beta^{(m)} + \left(K_{\beta\beta}^{(m)}\right)^{-1} U_\beta^{(m)}(\beta) \quad \text{and} \quad \gamma^{(m+1)} = \gamma^{(m)} + \left(K_{\gamma\gamma}^{(m)}\right)^{-1} U_\gamma^{(m)}(\gamma), \tag{13}$$

where $m = 0,1,2,\ldots$ indexes iterations that are performed until convergence is reached, which occurs when the distance between $\beta^{(m+1)}$ and $\beta^{(m)}$ becomes smaller than a predetermined small positive precision constant.

References

Barndorff-Nielsen, O. E., & Jorgensen, B. (1991). Some parametric models on the simplex. *Journal of Multivariate Analysis*, *39*(1), 106–116.

Belhadi, A., Djenouri, Y., Djenouri, D., Michalak, T., & Lin, C.-W. J. (2021). Machine learning for identifying group trajectory outliers. *ACM Transactions on Management Information Systems*, *12*(2), Article 12.

Brownlee, K. (1965). *Statistical Theory and Methodology in Science and Engineering*. New York: Wiley.

Cox, D. R., & Reid, N. (1987). Parameter orthogonality and approximate conditional inference. *Journal of the Royal Statistical Society B*, *49*(1), 1–39.

Dunn, P. K., & Smyth, G. K. (1996). Randomized quantile residuals. *Journal of Computational and Graphical Statistics*, *5*(3), 236–244.

Espinheira, P. L., Ferrari, S. L. P., & Cribari-Neto, F. (2008). Influence diagnostics in beta regression. *Computational Statistics & Data Analysis*, *52*(9), 4417–4431.

Espinheira, P. L., Ferrari, S. L. P., & Cribari-Neto, F. (2014). Bootstrap prediction intervals in beta regressions. *Computational Statistics*, *29*(5), 1263–1277, Erratum: 32, 2017, 1777.

Espinheira, P. L., Santos, E. G., & Cribari-Neto, F. (2017). On nonlinear beta regression residuals. *Biometrical Journal*, *59*(3), 445–461.

Espinheira, P. L., & Silva, A. O. (2020). Residual and influence analysis to a general class of simplex regression. *TEST*, *29*(2), 523–552.

Espinheira, P. L., Silva, L. C. M., Silva, A. O., & Ospina, R. (2019). Model selection criteria on beta regression for machine learning. *Machine Learning and Model Extraction*, *1*(1), 427–449.

Ferrari, S. L. P., & Cribari-Neto, F. (2004). Beta regression for modelling rates and proportions. *Journal of Applied Statistics*, *31*(7), 799–815.

Garcia Salazar, S. M. (2005). *Contribuicion al Estudio de la Reaccion de Decomposicion de la Zeolita Y em Presencia de Vapor de Agua y Vanadio* (Master's thesis), Universidad Nacional de Colombia.

Gupta, S. (2015). A regression modeling technique on data mining. *International Journal of Computer Applications*, *116*(9), 27–29.

Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques*. San Francisco: Morgan Kaufmann Publishers.

Jørgensen, B. (1997). *The Theory of Dispersion Models*. London: Chapman and Hall.

Lemonte, A. J., & Bazán, J. L. (2015). A new class of johnson S_B distribution and its associated regression model for rates and proportions. *Biometrical Journal*, *58*(4), 727–746.

Maniruzzaman, M., Rahman, M. J., Al-MehediHasan, M., Suri, H. S., Abedin, M. M., El-Baz, A., & Suri, J. S. (2018). Accurate diabetes risk stratification using machine learning: role of missing value and outliers. *Journal of Medical Systems*, *42*, Article 92.

Mitnik, P. A., & Baek, S. (2013). The kumaraswamy distribution: median-dispersion re-parameterizations for regression modeling and simulation-based estimation. *Statistical Papers*, *54*(1), 177–192.

Mousa, A. M., El-Sheikh, A. A., & Abdel-Fattah, M. A. (2016). A gamma regression for bounded continuous variables. *Advances and Applications in Statistics*, *49*(4), 305–326.

Ospina, R., Cribari-Neto, F., & Vasconcellos, K. L. P. (2006). Improved point and interval estimation for a beta regression model. *Computational Statistics & Data Analysis*, *51*(2), 960–981, Erratum: 55, 2011, 2445.

Pregibon, D. (1981). Logistic regression diagnostics. *The Annals of Statistics*, *9*(4), 705–724.

Qiu, Z., Song, P. X. K., & Tan, M. (2008). Simplex mixed effects models for longitudinal proportional data. *Scandinavian Journal of Statistics*, *35*(4), 577–596.

Rocha, A. V., & Cribari-Neto, F. (2009). Beta autoregressive moving average models. *TEST*, *18*, 529–545.

Rocha, A. V., & Cribari-Neto, F. (2017). Erratum to: beta autoregressive moving average models. *TEST*, *26*, 451–459.

Rousseeuw, P. J., & van Zomeren, B. C. (1990). Unmasking multivariate outliers and leverage points. *Journal of the American Statistical Association*, *85*(411), 633–639.

Schmid, M., Wickler, F., Maloney, K. O., Mitchell, R., Fenske, N., & Mayr, A. (2013). Boosted beta regression. *PLoS One*, *8*(4), Article e61623.

Smithson, M., & Shou, Y. (2017). Cdf-quantile distributions for modelling random variables on the unit interval. *British Journal of Mathematical and Statistical Psychology*, *70*(3), 412–438.

Song, P. X. K. (2009). Dispersion models in regression analysis. *Pakistan Journal of Statistics*, *25*(9), 529–551.

Song, P. X. K., Qiu, Z., & Tan, M. (2004). Modelling heterogeneous dispersion in marginal models for longitudinal proportional data. *Biometrical Journal*, *56*(2), 540–553.

Song, P. X. K., & Tan, M. (2000). Marginal models for longitudinal continuous proportional data. *Biometrics*, *56*(2), 496–502.

Weiss, S. M., & Indurkhya, N. (1997). *Predictive Data Mining: A Practical Guide*. San Francisco: Morgan Kaufmann Publishers.

Weizhi, L., M., W., Zhang, X., J.J., S., Lu, Y., Sellke, E. W., Fan, W., DiMaio, J. M., & Thatcher, J. (2015). Outlier detection and removal improves accuracy of machine learning approach to multispectral burn diagnostic imaging. *Journal of Biomedical Optics*, *20*(12), 1–9.

Zhang, P., Qiu, Z., & Shi, C. (2016). Simplexreg: an r package for regression analysis of proportional data using the simplex distribution. *Journal of Statistical Software*, *71*(11), 1–21.